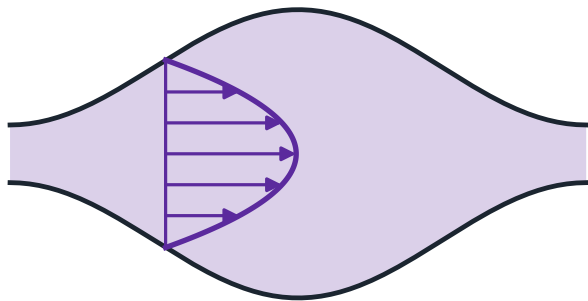


Analytical approximation

Fast reduced model

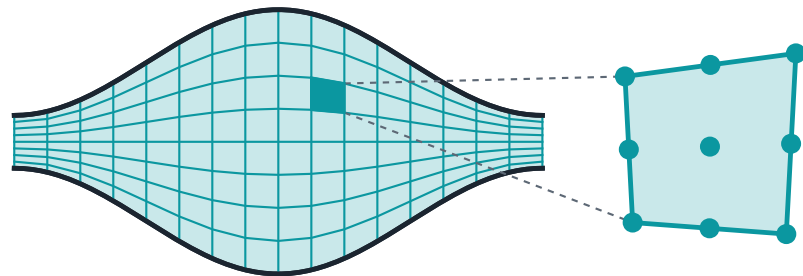


Lubrication approximation

Closed-form flow-rate estimate

Finite-element method

Accurate numerical reference

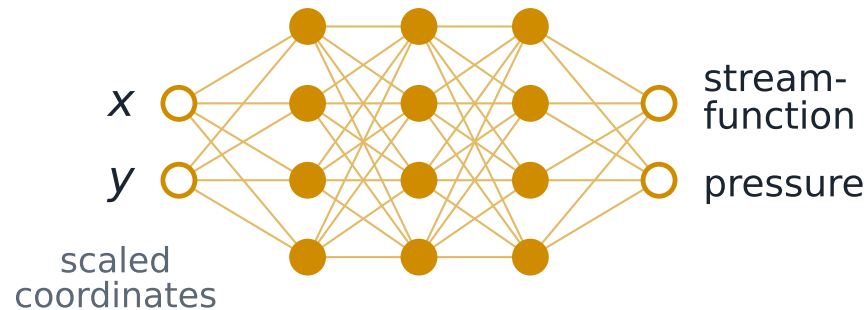


Full 2D Taylor-Hood solution

Periodic in x, no-slip walls

Physics-informed neural network

Physics-only neural solution



Governing equations + boundary conditions

No FEM or analytical training data